



ISTITUTO NAZIONALE DI GEOFISICA E VULCANOLOGIA



CoARA

CoARA Action Plan

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Istituto Nazionale di Geofisica e Vulcanologia (INGV)

Introduction

The Istituto Nazionale di Geofisica e Vulcanologia (INGV) joined CoARA in July 2023.

The CoARA INGV working group has agreed to proceed with the preparation of a draft action plan, initially based on the guidance provided by CoARA. This includes support for CoARA signatories in the preparation of action plans and guidance on how to approve action plans internally and how to share them with the broader community. The CoARA community has made available on Zenodo action plans shared by other institutions and reflections made for the alignment of roadmaps under Task 1.2 of the Italian National Chapter in the document "Guiding Questions." Additionally, insights from the ongoing work of working groups and meetings have been incorporated. This version served for discussion with INGV staff at the INGV Symposium (November 11-15, 2024) . After being submitted for approval to the Board of Advisors (CdA Consiglio di Amministrazione), it will be published in the Zenodo directory.

Starting point

Reflect on your strategy and change approach

INGV signed the Agreement on Reforming Research Assessment (ARRA) in July 2023, following the Board of Advisors resolution of 28/4/2023, aiming to reinforce the value of the diversity of products, results, and practices that constitute the geoscientific research carried out at INGV.

By signing these documents, INGV supports a global reform of the scientific endeavour, that moves from purely competitive to collaborative and more inclusive.

To coordinate and carry out the activities related to the fulfillment of the actions resulting from the signing of the Agreement, INGV set up a dedicated working group, the Gruppo di Lavoro (GdL) CoARA INGV. The GdL consists of researchers and technologists with different expertise, including a representative of each Section in which INGV is composed, and it is coordinated by the responsible for the Office where the periodical evaluation of research is performed (Ufficio per la valorizzazione e la valutazione della Ricerca) of the Central Administration.

As a first step, it was agreed to focus on principle 1 "Recognize the diversity of contributions and careers in research, depending on the needs and nature of the research itself."

Involve your institutional community in the change process

The GdL organized meetings with top management and staff and took part in the activities of the Italian National Chapter and of thematic international Working Groups to benefit from mutual learning and networking activities.

It was decided to undertake the reform journey with a participatory approach, evolving gradually and in a shared way, organizing staff assemblies regularly (bimonthly) to discuss, identify, and

propose/follow a path of real and substantial reform, engaging various stakeholders both in the research team and the administrative structures.

The CoARA roadmap will be shared with the entire INGV community through ad hoc meetings (e.g. INGV Annual Symposium), email communication, and publication on the website.

Identify key challenges to address

The evaluation of research with current criteria shows limitations:

- It produces “adaptive” behaviors to variations in evaluation parameters, possibly deteriorating the scientific value of the results achieved;
- It promotes competition, not maximizing the scientific results at the expense of more productive forms of collaboration;
- It fuels the ineffective system of publishing in journals deemed “prestigious,” with high impact factors, paid at a high price;
- It does not recognize important contributions to research, such as: service, monitoring networks implementation and management, research infrastructures, data storage, software, communication and dissemination;
- It does not recognize the multiplicity and diversity of contributions in the evaluation of the Institute, nor in that of Researcher/Technologist/Technician personnel;
- It encourages publication in “predatory” and “hijacked” journals, which ensure much faster publication times;
- It favors the emergence of more and more paid open access journals, a harmful distortion of the concept of open access;
- It has a distorting and deteriorating impact on the peer review process, which represents the fundamental certification of scientific progress (also as a consequence of the editorial dynamics mentioned above);
- It markedly tends to homogenize the scientific questions to address, as it discourages “risky” or unpopular, non-mainstream research, less profitable and less likely to pass the peer review process;
- It discourages the publication of negative results (which in the current conception are often not even considered results themselves), causing a waste of time and money for those who have to retrace the same paths without knowing.

A thorough reform of the research system should be imagined, that not only concerns evaluation of research products and practices, but it also addresses the other pillars of the present dysfunctional research system: peer review, currently inefficient and affected by systemic biases; and the purely competitive research funding mechanism based on proposals evaluation, which, on one hand, are increasingly contaminated by transient scientific trends and, on the other, suffer from basic fundamental limitations.

INGV will promote and foster open and free discussion among researchers in order to increase awareness and direction on these fundamental issues and help the identification and development of an actual reforming path.

Operational Action Plan - Core Commitments

Recognise the diversity of contributions to, and careers in, research in accordance with the needs and nature of the research

INGV aims at including into its research assessment practices, including funding distribution and promotion, more research products than journal articles only. Products that can be included in evaluation procedures are:

- software
- datasets
- peer reviews
- research infrastructures
- outreach and teaching material
- patents
- geological maps.

INGV will require, dedicating adequate funding, that output from research projects is open and shared in FAIR repositories, unless some confidentiality is needed (“as open as possible, as closed as necessary”).

Avoid the use of rankings of research organization in research assessment

There is a general consensus that rankings should not be used in research evaluation. However, the conversion of this consensus into an abandonment of rankings would entail a significant cultural shift and would therefore be a challenging undertaking.

Responsible research evaluation should promote expert judgment and never rely on research metrics only to assess excellence. Despite the claims of many national institutions and research groups, recent studies have shown that the correlation between journal rankings and expert judgements has not changed. Furthermore, this correlation is stronger for the most prestigious journals. Additionally, the correlation has not changed for institutions that have adhered to the DORA principles.

Commit resources to reforming research assessment as is needed to achieve the organisational changes committed to

INGV is committed to providing the necessary financial and human resources to implement training and awareness initiatives that will facilitate the organisational changes required to implement the reform.

Review and develop research assessment criteria, tools and processes

Career Evaluation. INGV will shift towards narrative CVs in career progression competitions.

A CV of this type would allow researchers to outline their profile based on three key dimensions: advancing research frontiers, enhancing the results of research for the scientific, cultural, technological, and economic development of the country, and providing technical-scientific support through dissemination and communication activities.

The calls themselves should focus on career paths, giving priority to both research products and management aspects, allowing researchers to influence the evaluation of their professional path. This way, researchers and technologists become active protagonists in their career development. To promote the sharing of results and collaboration among researchers, the narrative CV could include case studies on how early data sharing and collaboration efforts led to the creation of knowledge and results that otherwise would not have been achieved.

Appropriate and in-depth reflections are necessary to identify possible corrective mechanisms. In order to try to limit the possible dispersive effects of the approach described above and to identify a mechanism also “rewarding vertical excellence”.

All products will be evaluated on the basis of qualitative parameters, aided by responsible use of quantitative metrics. Adoption of potentially biasing and non-virtuous procedures like scores saturation (especially micro segmented multiple scores saturation) has to be avoided. To mitigate the effects of a purely linearly normalized score system, alternative normalization curves could be designed and optionally adopted.

Raise awareness of research assessment reform and provide transparent communication, guidance, and training on assessment criteria and processes as well as their use

INGV plans to provide guidelines for the hiring and promotion commissions in order to promote excellence in the framework of an inclusive evaluation of the research.

INGV will foster discussion on research evaluation best practices through recurrent internal meetings for the whole personnel, to gather community input and reflections.

INGV commits to raise awareness about unconscious bias and organize training for evaluators benefitting from guidelines and products collected in the CoARA WG “TIER-Towards an Inclusive Evaluation of Research”.

Appropriate reflections should be done about the composition of the evaluation commissions (internal and external).

Communicate progress made on adherence to the principles and implementation of the Commitments

INGV is committed to providing updates on the progress of the action plan in accordance with the timeline outlined in the Research Assessment Reform Agreement.

INGV will communicate progress through the publication of relevant documents and reports on the INGV COARA website and on the Zenodo platform or similar.

Evaluate practices, criteria and tools based on solid evidence and the state-of-the-art in research on research, and make data openly available for evidence gathering and research

The CoARA roadmap for INGV will be subject to a review within the second year of application (2027).